

Fitness Club Database

Final Project — Database Modeling Document

This document presents the full database design for a **Fitness Club Management System**. It covers the conceptual entity-relationship overview, a complete logical schema with all columns, data types, constraints, and key roles, and a 3NF justification for each table.

Database: fitness_club | **Schema:** fitness_club_db

1.1 Domain Description

The system manages the day-to-day operations of a fitness club. Members register with a chosen membership plan, instructors lead workout classes that are scheduled in specific facilities, and members enrol into those sessions. Every financial transaction is recorded as a payment linked to the responsible member.

- Members purchase one of several membership tiers (Gold, Silver, Bronze, Platinum, Trial).
- Instructors have a defined specialization (Yoga, Cardio, MMA, Gym, Pilates) and an hourly rate.
- Classes are categorised (Yoga, Cardio, Strength, Combat, Wellness) and scheduled in a facility on a specific date.
- The many-to-many relationship between Members and Schedule is resolved via the Schedule_Enrollment junction table.
- Payments are immutable financial records; once inserted they cannot be updated (enforced via REVOKE).

1.2 Conceptual ERD — Relationships

Entity A	Cardinality	Entity B	Relationship Label
membership_types	1 : N	members	A member holds one membership type
members	1 : N	schedule_enrollment	A member can enrol in many sessions
schedule	1 : N	schedule_enrollment	A session can have many enrolments
members	N : M	schedule	Resolved via schedule_enrollment (junction)
classes	1 : N	schedule	A class can be scheduled many times
instructors	1 : N	schedule	An instructor leads many sessions
facilities	1 : N	schedule	A facility hosts many sessions
members	1 : N	payments	A member can make many payments

1.3 Logical Schema

Each table is described with its column name, data type, nullability, key role, and any constraints applied.

Table: membership_types

Column	Data Type	Nullable	Key / Role	Constraints / Notes
membership_type_id	SERIAL	NOT NULL	PK	Auto-increment primary key
name	VARCHAR(50)	NOT NULL	UNIQUE	Natural key; membership tier name
duration_months	INT	NOT NULL		CHECK (duration_months > 0)
price	NUMERIC(10,2)	NOT NULL		CHECK (price >= 0)
access_level	VARCHAR(50)	NOT NULL		CHECK IN ('VIP','Standard','Basic')
is_active	BOOLEAN	NOT NULL	DEFAULT	DEFAULT true
record_ts	DATE	NOT NULL	DEFAULT	DEFAULT CURRENT_DATE

Table: members

Column	Data Type	Nullable	Key / Role	Constraints / Notes
member_id	SERIAL	NOT NULL	PK	Auto-increment primary key
first_name	VARCHAR(50)	NOT NULL		
last_name	VARCHAR(50)	NOT NULL		
email	VARCHAR(100)	NOT NULL	UNIQUE	Natural key; must be unique
phone	VARCHAR(25)	NOT NULL		NOT NULL — contact required
date_of_birth	DATE	NOT NULL		
membership_type_id	INT	NOT NULL	FK	→ membership_types; ON DELETE RESTRICT
joined_date	DATE	NOT NULL		CHECK (joined_date > DATE '2026-01-01')
emergency_contact	VARCHAR(20)	NULL		Added via ALTER TABLE
record_ts	DATE	NOT NULL	DEFAULT	DEFAULT CURRENT_DATE

Table: instructors

Column	Data Type	Nullable	Key / Role	Constraints / Notes
instructor_id	SERIAL	NOT NULL	PK	Auto-increment primary key
first_name	VARCHAR(50)	NOT NULL		
last_name	VARCHAR(50)	NOT NULL		
specialization	VARCHAR(50)	NOT NULL		CHECK IN ('Yoga','Cardio','MMA','Gym','Pilates')
hourly_rate	NUMERIC(10,2)	NOT NULL		CHECK (hourly_rate >= 0)
rating	NUMERIC(3,1)	NULL	DEFAULT	DEFAULT 5.0 — added via ALTER TABLE
record_ts	DATE	NOT NULL	DEFAULT	DEFAULT CURRENT_DATE

Table: classes

Column	Data Type	Nullable	Key / Role	Constraints / Notes
class_id	SERIAL	NOT NULL	PK	Auto-increment primary key
name	VARCHAR(100)	NOT NULL	UNIQUE	Natural key
description	TEXT	NULL		
category	VARCHAR(50)	NOT NULL		CHECK IN ('Yoga','Cardio','Strength','Combat','Wellness')
record_ts	DATE	NOT NULL	DEFAULT	DEFAULT CURRENT_DATE

Table: facilities

Column	Data Type	Nullable	Key / Role	Constraints / Notes
facility_id	SERIAL	NOT NULL	PK	Auto-increment primary key
name	VARCHAR(50)	NOT NULL		
capacity	INT	NOT NULL		CHECK (capacity >= 0)
location_description	VARCHAR(100)	NULL		
record_ts	DATE	NOT NULL	DEFAULT	DEFAULT CURRENT_DATE

Table: schedule

Column	Data Type	Nullable	Key / Role	Constraints / Notes
schedule_id	SERIAL	NOT NULL	PK	Auto-increment primary key
class_id	INT	NOT NULL	FK	→ classes; ON DELETE RESTRICT
instructor_id	INT	NOT NULL	FK	→ instructors; ON DELETE RESTRICT
facility_id	INT	NOT NULL	FK	→ facilities; ON DELETE RESTRICT
session_date	DATE	NOT NULL		CHECK (session_date > DATE '2026-01-01')
duration_minutes	INT	NOT NULL		CHECK (duration_minutes > 0)
duration_hours	NUMERIC(5,2)	NOT NULL	GENERATED	GENERATED ALWAYS AS (duration_minutes/60.0) STORED
max_participants	INT	NOT NULL	DEFAULT	DEFAULT 20 — added via ALTER TABLE
record_ts	DATE	NOT NULL	DEFAULT	DEFAULT CURRENT_DATE

Table: schedule_enrollment (junction — resolves Members ↔ Schedule M:N)

Column	Data Type	Nullable	Key / Role	Constraints / Notes
member_id	INT	NOT NULL	PK + FK	→ members; ON DELETE CASCADE; part of composite PK
schedule_id	INT	NOT NULL	PK + FK	→ schedule; ON DELETE CASCADE; part of composite PK
enrolled_at	TIMESTAMP	NOT NULL	DEFAULT	DEFAULT CURRENT_TIMESTAMP
record_ts	DATE	NOT NULL	DEFAULT	DEFAULT CURRENT_DATE

Table: payments

Column	Data Type	Nullable	Key / Role	Constraints / Notes
payment_id	SERIAL	NOT NULL	PK	Auto-increment primary key
amount	NUMERIC(10,2)	NOT NULL		CHECK (amount >= 0)
payment_date	TIMESTAMP	NOT NULL	DEFAULT	DEFAULT CURRENT_TIMESTAMP; CHECK > '2026-01-01 00:00:00'
payment_method	VARCHAR(30)	NOT NULL		CHECK IN ('Cash','Card','Transfer')
member_id	INT	NOT NULL	FK	→ members; ON DELETE RESTRICT
payment_note	TEXT	NULL		Renamed from 'note' via ALTER TABLE
record_ts	DATE	NOT NULL	DEFAULT	DEFAULT CURRENT_DATE